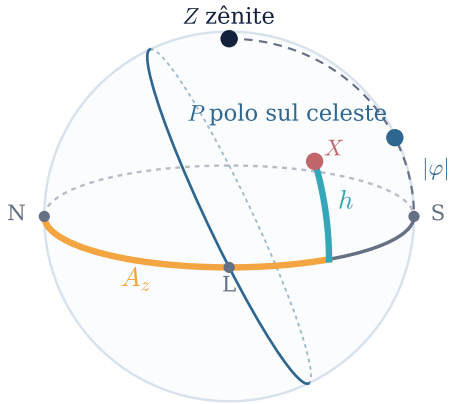
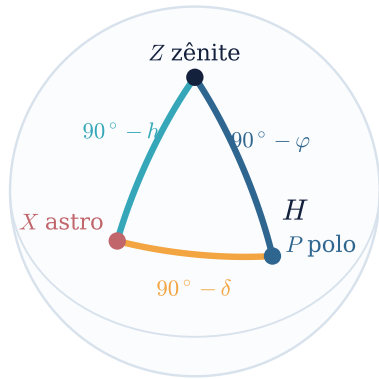


(a) a esfera celeste em Joinville



azimute contado do Norte para Leste;
altura medida sobre o círculo vertical

(b) o triângulo esférico PZX



$$\sin h = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos H$$